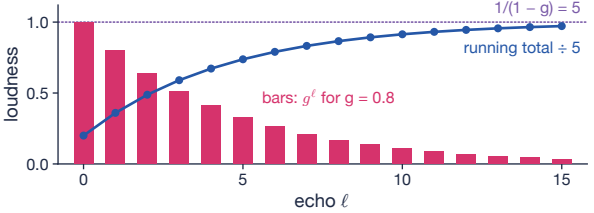
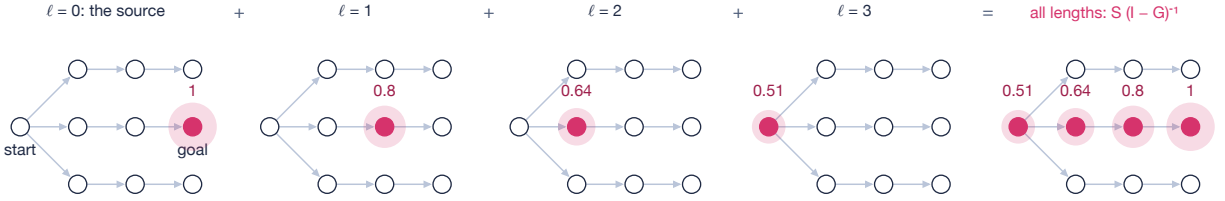
a Echoes: each bounce quieter by a factor g



$1 + g + g^2 + g^3 + \ldots = 1/(1 - g)$ when $|g| < 1$

For a matrix G the same sum over powers is $I + G + G^2 + \ldots = (I - G)^{-1}$, the resolvent. It converges when repeated multiplication by G shrinks every vector (the spectral radius of G is below one).

b The same sum over walks: light flowing back from the goal (gate 0.8 on each reversed road)



Each power of G carries the light one more road back, against the arrows. The total lights only the start's child that leads to the goal; a walk of ℓ roads keeps weight 0.8^ℓ.